Centre for Biomedical Technologies

MODELLING THE BIOMECHANICAL PROPERTIES OF THE PLACENTA USING SONOELASTOGRAPHY

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Queensland family COHORT STUDY

mater research

Placentation

Cellular

Maternal cardiovascular

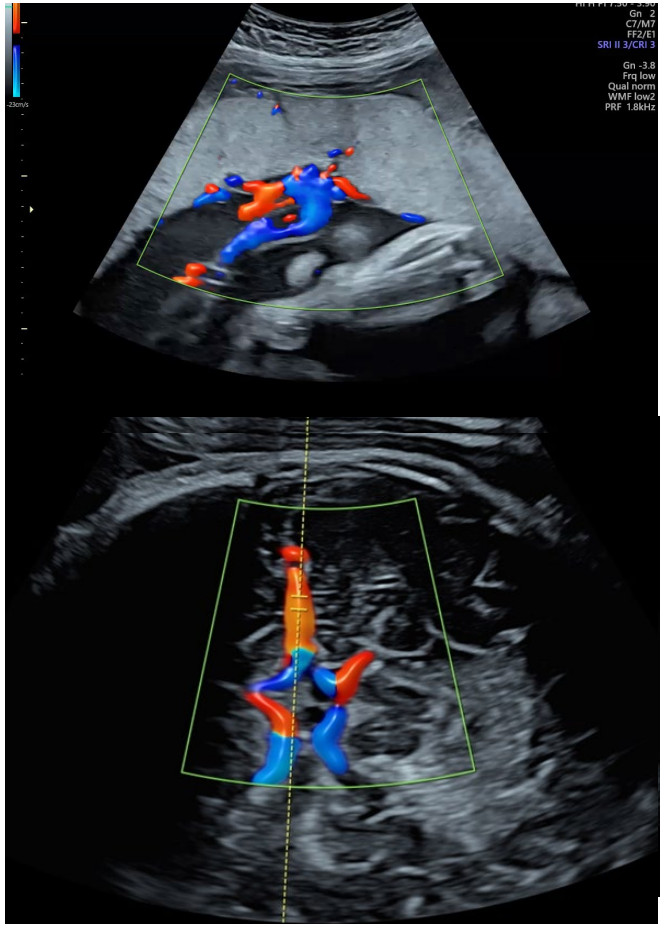
Growth Restriction

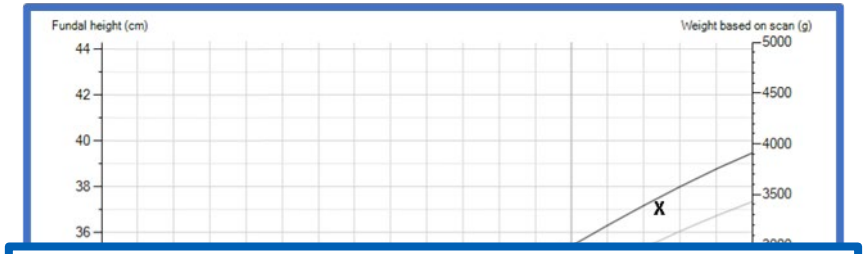
Preeclampsia

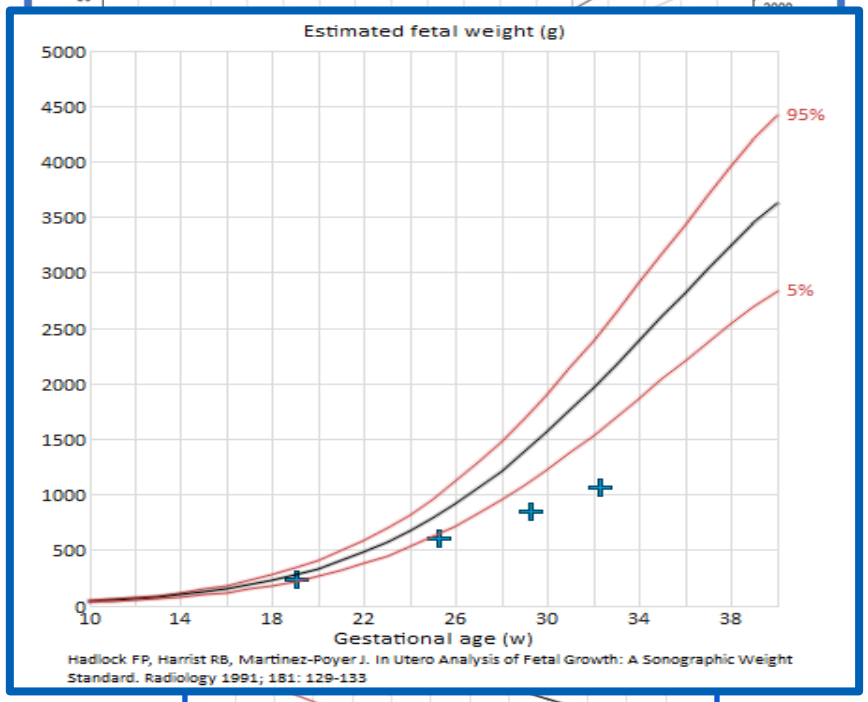
Gestational Diabetes

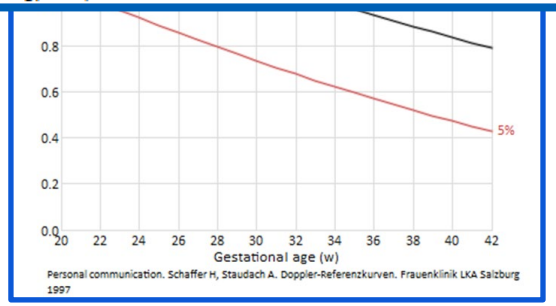
Biometry

Doppler









EARLY FGR

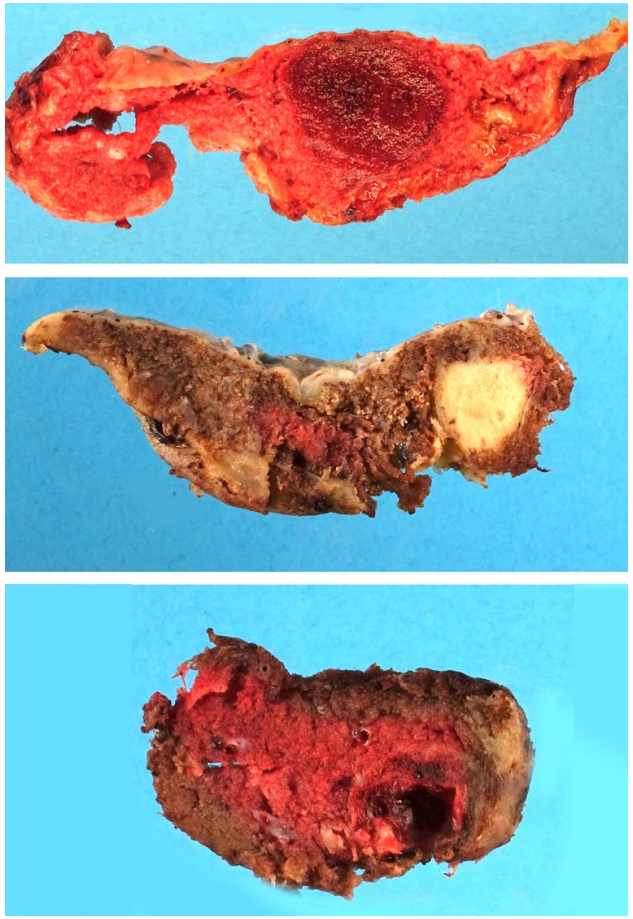
- Prior to 32-34 weeks' gestation
- Severe placental disease
- Severe FGR (<1st centile)
- Abnormal UA Doppler
- High assoc. with PET
- High morbidity and mortality

LATE FGR

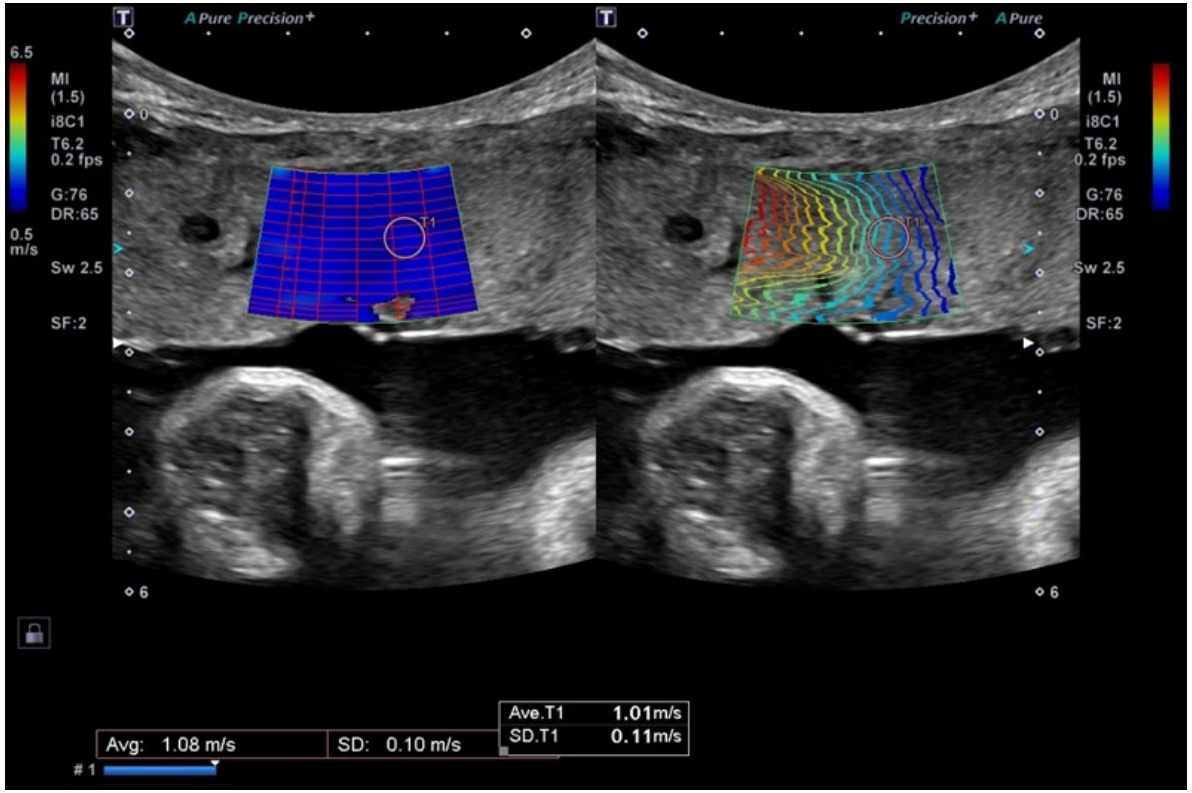
- After 34 weeks' gestation
- Mild placental disease
- Not always SGA
- UA often normal
- Lower assoc. with PET
- Lower morbidity BUT
- **COMMON CAUSE OF LATE STILLBIRTH**

What if we could measure these changes directly within the placenta?

What is the normal progression of placental stiffness throughout late pregnancy?



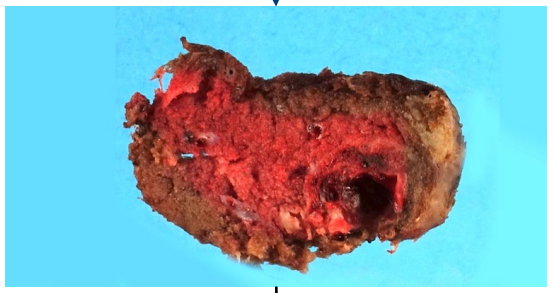
Shear Wave Elastography (SWE)



Shear wave velocity (SWV) ∝ elasticity

Defective placentation

Infarction / fibrosis / ischaemia



↑ SWV

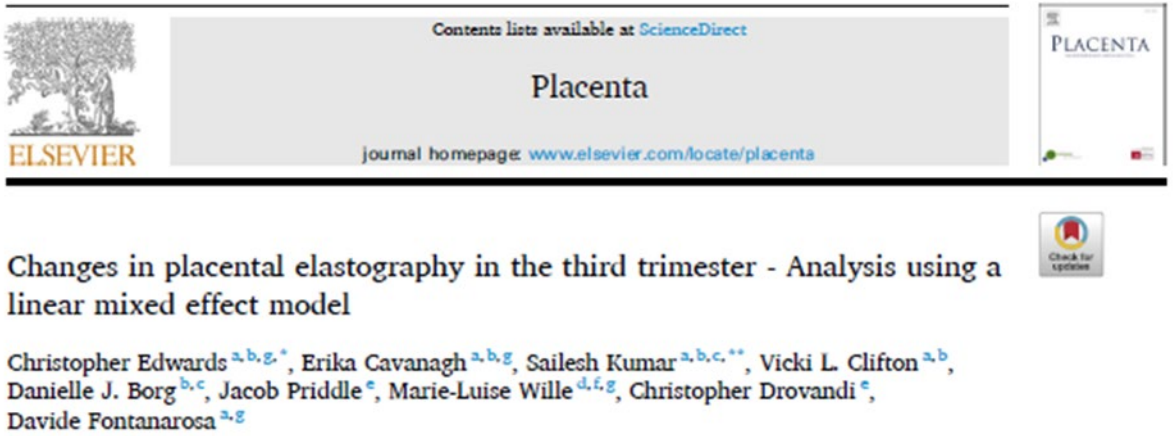
Model

238 Women with singleton pregnancy, recruited as part of the QLD Family Cohort

- Fetal Biometry (24, 28, 36 weeks)
- Amniotic Fluid
- Doppler Indexes
- Shear Wave Elastography
- Maternal biological and physical measurements

Linear mixed effects model

Predictors for SWV

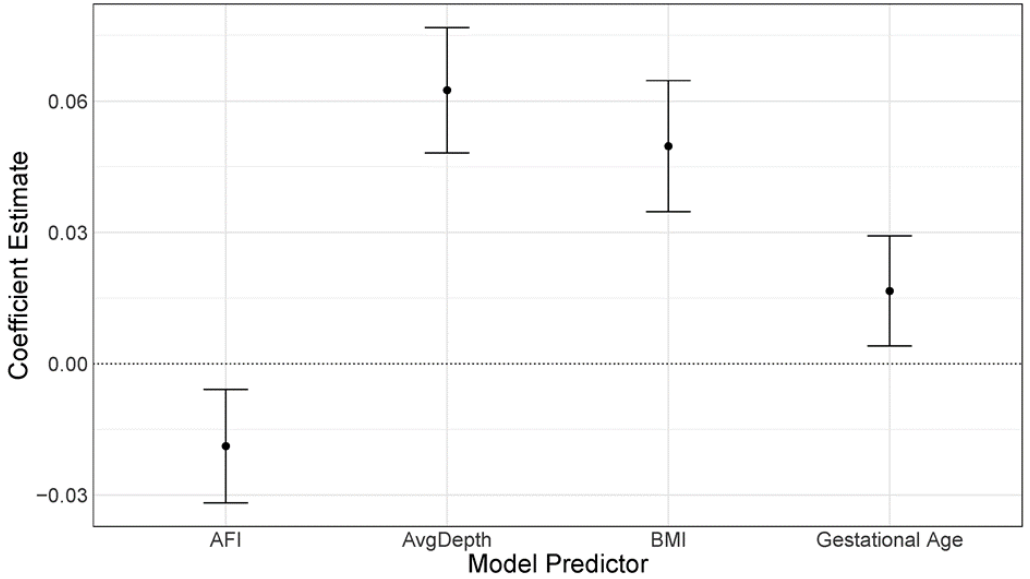

Contents lists available at ScienceDirect

Placenta

journal homepage: www.elsevier.com/locate/placenta

Changes in placental elastography in the third trimester - Analysis using a linear mixed effect model

Christopher Edwards [a,b,g,*], Erika Cavanagh [a,b,g], Sailesh Kumar [a,b,c,**], Vicki L. Clifton [a,b], Danielle J. Borg [b,c], Jacob Priddle [e], Marie-Luise Wille [d,f,g], Christopher Drovandi [e], Davide Fontanarosa [a,g]



$$Mean_SWV_{ij} = \beta_0 + \beta_1 \times Gestational\ Age_{ij} + \beta_2 \times AvgDepth_{ij} + \beta_3 \times BMI_{ij} + \beta_4 \times AFI_{ij} + \gamma_i + \epsilon_{ij}$$

Amniotic Fluid Index
Sampling Depth
Body Mass Index (BMI)
Gestational age

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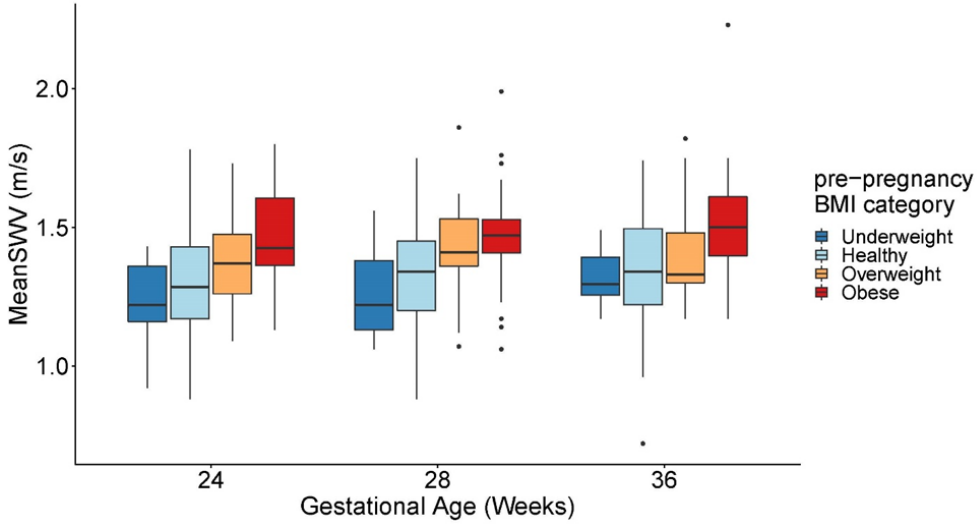
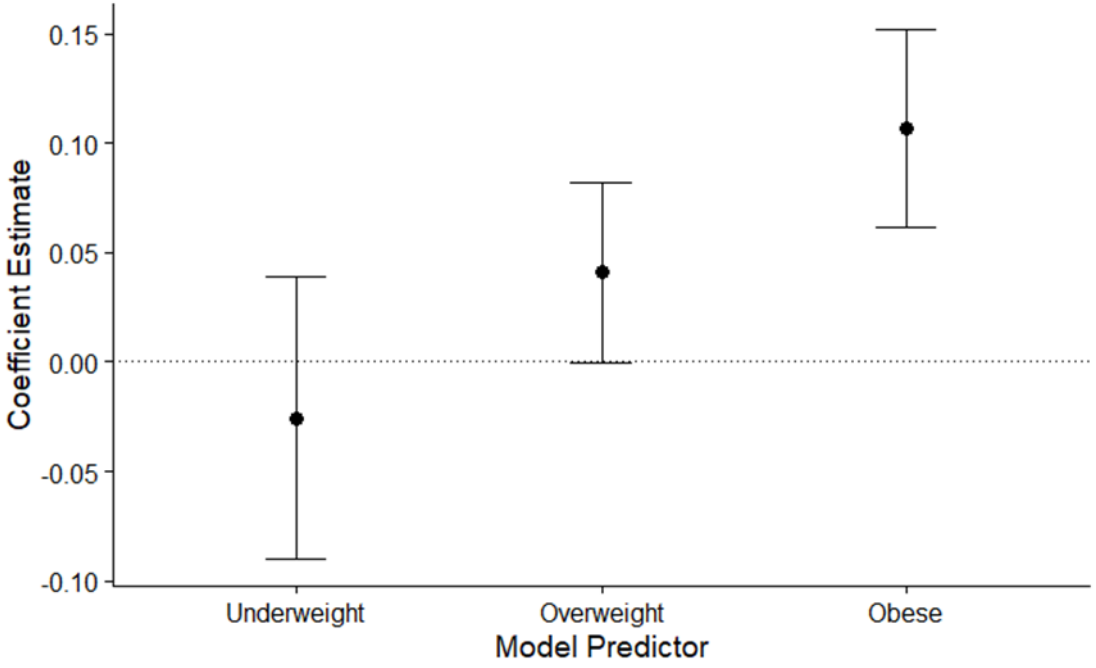
Relationship between placental elastography, maternal pre-pregnancy body mass index and gestational weight gain

Christopher Edwards [a,b,g,**], Erika Cavanagh [a,b,g], Sailesh Kumar [a,b,c,*], Vicki L. Clifton [b,c], Danielle J. Borg [b,c], Jacob Priddle [e,h], Wille Marie-Luise [d,f,g], Christopher Drovandi [e,h], Davide Fontanarosa [a,g]





Final thoughts

- Baseline model of normal placental biomechanical properties.
- Tool to monitor interventions
- Linked maternal obesity <> placental stiffness
- Greater potential to develop better diagnostic tools /
- Potential to combine sonoelastography with Biometry / Doppler / placental volume to make better predictions.

Trial begins of AI scan that could reduce risk of stillbirth and other conditions

Exclusive: scanning technique using AI to analyse ultrasound images could help assess risk of adverse pregnancy outcomes

https://www.theguardian.com/science/2022/mar/15/placenta-analysis-trial-seeks-to-identify-adverse-pregnancy-outcome-risk

QUT Centre for Biomedical Technologies

Thanks…

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mater research

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Prof. Vicki Clifton

Dr Danielle Borg

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